

Male Fertility Classification

Machine Learning Report

HARTAJ BAJWA SINGH

hartaj.ca@gmail.com

Abstract

This report describes the design, methodological choices, and evaluation of a binary classification system for male fertility diagnosis, based on the `fertility_Diagnosis.txt` dataset.

The objective is to predict whether a subject presents a normal (N) or altered (O) diagnosis from 9 input features. Given the severe class imbalance (approximately 88% normal, 12% altered), specific strategies were adopted to prevent the classifier from being trivially biased towards the majority class.

Four classification models were compared: Logistic Regression, Support Vector Machine (SVM), Random Forest, and Multilayer Perceptron (MLP). The SVM achieved the best performance in cross-validation and was therefore selected for the final evaluation on the test set, where it attained a Balanced Accuracy of 0.644.

1 Task Definition

The goal of this project is to build a classification system for male fertility diagnosis. The problem is formulated as a binary classification task, in which each subject must be assigned to one of two classes:

- N (Normal): normal fertility;
- O (Altered): altered fertility.

Classification is performed from a set of provided features. The dataset exhibits a strong imbalance between the two output values; consequently, specific choices were made regarding the evaluation metric and the train/test split strategy, in order to avoid models that systematically favour the majority class N.

2 Dataset and Data Preparation

The dataset consists of 100 samples with 9 features and a binary target variable. The features encompass variables related to the period of the analysis, lifestyle habits, and patient-specific information.

2.1 Feature Description

The 9 features used for classification are listed in Table 1.

Feature	Description	Admissible Values
Season	Season in which the analysis was performed.	Winter = -1, Spring = -0.33, Summer = 0.33, Fall = 1
Age	Patient age at the time of analysis.	Normalised value in $[0, 1]$ (actual age: 18–36 years)
Childish diseases	History of childhood diseases.	No = 0, Yes = 1
Accident or trauma	Presence of accidents or traumas.	No = 0, Yes = 1
Surgical intervention	History of surgical procedures.	No = 0, Yes = 1
High fevers	Episodes of high fever in the last 3 months.	< 3 months = -1, > 3 months = 0, No fever = 1
Alcohol consumption	Frequency of alcohol consumption.	Normalised value in $[0, 1]$ (from “several times a day” to “never”)
Smoking habit	Smoking behaviour.	Never = -1, Occasional = 0, Daily = 1
Sitting hours	Number of hours spent sitting per day.	Normalised value in $[0, 1]$ (actual range: 0–16 hours)

Table 1: Description of the features in the Fertility dataset.

2.2 Class Imbalance

As noted above, approximately 88% of the samples belong to class N and only 12% to class 0. This is a critical aspect, since a classifier that always predicts N would already achieve 88% accuracy, while being entirely uninformative — particularly in a medical context. To address this issue:

- `stratify=y` is used in `train_test_split` to preserve the class proportions in both the training and test sets;
- **Balanced Accuracy** is adopted as the primary evaluation metric, which computes the arithmetic mean of per-class recall and therefore assigns equal weight to both classes regardless of their prevalence;
- `class_weight='balanced'` is set in models that support it, so that misclassification errors on the minority class 0 are penalised more heavily.

2.3 Dataset Split

The dataset is partitioned with a 75%/25% ratio, yielding 75 training samples and 25 test samples. This choice was reached after empirical evaluation, as discussed in Section 6. The split is fixed via `random_state=0` to ensure reproducibility across runs.

2.4 Feature Scaling

Features are standardised using `StandardScaler` fitted on the training set. The same scaler is subsequently applied to the test set, thereby preventing any form of data leakage. Scaling is required by all models considered, with the exception of Random Forest.

2.5 Correlation Analysis

To identify potential linear relationships among variables, the correlation matrix shown in Figure 1 was computed. The analysis reveals no notably high linear correlations between any pair of features.

Figure 1: Correlation matrix of the features.

3 Models and Hyperparameter Selection

Hyperparameter selection is carried out via **Grid Search with 5-fold cross-validation** (`GridSearchCV`), optimising Balanced Accuracy. The motivation behind the choice of each model and its associated hyperparameter grid is detailed below.

3.1 Logistic Regression

Logistic Regression was selected because of its suitability for small datasets, where more complex models tend to overfit. It natively supports `class_weight='balanced'`.

The hyperparameter grid is:

- **penalty**: 11 or 12. L1 regularisation drives some weights to exactly zero, which is useful for implicit feature selection, whereas L2 distributes the penalty uniformly across all weights.
- **C** $\in \{10^{-5}, 5 \times 10^{-5}, 10^{-4}, 5 \times 10^{-4}, 1\}$: the inverse of the regularisation strength. Small values of **C** correspond to stronger regularisation.

Best configuration found: C=1e-05, penalty=12. Balanced Accuracy (CV): 0.564.

3.2 Support Vector Machine (SVM)

The SVM was included because it supports both linear and non-linear decision boundaries through the use of different kernels, providing greater modelling flexibility. It natively supports `class_weight='balanced'`.

The hyperparameter grid is:

- **C** $\in \{0.1, 1, 10, 100\}$: controls the trade-off between the margin width and the number of classification errors.
- **gamma** $\in \{\text{scale}, \text{auto}, 0.01, 0.1\}$: defines the influence of each individual training sample on the decision boundary when using the RBF kernel. The **scale** and **auto** options adapt to the data scale.
- **kernel**: **linear** (linear classifier) and **rbf** (non-linear classifier via Gaussian kernel).

Best configuration found: C=1, gamma=auto, kernel=rbf. Balanced Accuracy (CV): 0.624.

3.3 Random Forest

Random Forest was selected to evaluate an ensemble tree-based approach. The method is robust to noise and capable of modelling non-linear relationships without requiring distributional assumptions about the data. It natively supports `class_weight='balanced'`.

The hyperparameter grid is:

- **n_estimators** $\in \{50, 100, 150\}$: number of trees in the forest. A larger ensemble reduces variance at the cost of increased computational time.

- `max_depth` $\in \{3, 5, 7, \text{None}\}$: maximum depth of each tree. Constraining depth reduces overfitting, which is particularly relevant for small datasets.
- `min_samples_split` $\in \{2, 5\}$: minimum number of samples required to split an internal node.

Best configuration found: `max_depth=7`, `min_samples_split=2`, `n_estimators=50`. **Balanced Accuracy (CV): 0.540.**

3.4 Neural Network (MLP)

The Multilayer Perceptron was included to assess whether a neural network architecture could yield superior performance. A single hidden layer was used, as it is generally sufficient for datasets of this complexity.

The hyperparameter grid is:

- `hidden_layer_sizes` $\in \{n, n/2, 2n\}$ (where $n = 9$ is the number of features): configurations with a number of neurons equal to, half, or double the number of input features are explored.
- `alpha` $\in \{0.0001, 0.001, 0.01\}$: L2 regularisation coefficient applied to the network weights.
- `learning_rate_init` $\in \{0.001, 0.01, 0.1\}$: initial step size of the optimiser. The value 0.0001 was excluded as it leads to unnecessarily slow convergence on datasets of this size.

Best configuration found: `alpha=0.0001`, `hidden_layer_sizes=9`, `learning_rate_init=0.01`. **Balanced Accuracy (CV): 0.597.**

4 Final Model Selection

A summary of the mean Balanced Accuracy values obtained in cross-validation is provided in Table 2.

Model	Balanced Accuracy (CV)
SVM	0.624
MLP	0.597
Logistic Regression	0.564
Random Forest	0.540

Table 2: Mean Balanced Accuracy in cross-validation for each model.

The best-performing model is the **Support Vector Machine (SVM)**.

The optimal configuration identified by the Grid Search is:

- `C = 1`
- `gamma = auto`
- `kernel = rbf`

It achieved the highest mean Balanced Accuracy on the training set (0.624), making it the most suitable choice for the final testing phase.

Class	Precision	Recall	F1-score	Support
N (normal)	0.91	0.95	0.93	22
O (altered)	0.50	0.33	0.40	3
Macro avg	0.71	0.64	0.67	25
Weighted avg	0.86	0.88	0.87	25

Table 3: Classification report on the test set (`random_state=0`).

5 Test Set Results

After retraining the selected model on the entire training set, evaluation on the test set (25 samples) produced the following results:

The confusion matrix is:

$$\begin{pmatrix} 21 & 1 \\ 2 & 1 \end{pmatrix}$$

where rows represent the true classes (N, O) and columns represent the predicted classes.

5.1 Analysis of Results

- **Standard Accuracy: 0.88** — high, but must be interpreted with caution given the strong prevalence of class N in the test set (22 out of 25 samples).
- **Balanced Accuracy: 0.644** — confirms that the model is capable of distinguishing between both classes in a non-trivial manner, beyond simple class prevalence.
- **Recall on class O: 0.33** — the model correctly identifies 1 out of 3 samples belonging to the minority class.
- **Precision on class O: 0.50** — out of 2 positive predictions for class O, one is correct, reflecting a relatively low number of false positives for this split.

The limited performance on the minority class is primarily attributable to the small dataset size and the high degree of class imbalance. A larger and more balanced dataset would allow for a more reliable model estimation.

6 Sensitivity Analysis on Test Set Size

Given the limited size of the dataset, an exploratory analysis was conducted to assess how the choice of the `test_size` parameter influences the final evaluation metrics. All experiments were performed with `random_state=0` to isolate the sole effect of `test_size`. Three values were tested: 0.20, 0.25, and 0.30. In all cases, the selected model remained the Support Vector Machine.

test_size	Test samples	O in test	Accuracy	Bal. Acc.	Recall O	Precision O
0.20	20	2	0.90	0.722	0.50	0.50
0.25	25	3	0.88	0.644	0.33	0.50
0.30	30	4	0.87	0.606	0.25	0.50

Table 4: Comparison of evaluation metrics as a function of `test_size` (`random_state=0`).

The confusion matrices for the three cases are:

$$\text{test_size} = 0.20 : \begin{pmatrix} 17 & 1 \\ 1 & 1 \end{pmatrix} \quad \text{test_size} = 0.25 : \begin{pmatrix} 21 & 1 \\ 2 & 1 \end{pmatrix} \quad \text{test_size} = 0.30 : \begin{pmatrix} 25 & 1 \\ 3 & 1 \end{pmatrix}$$

6.1 Interpretation

The results show that as the test set size increases, Balanced Accuracy decreases monotonically, while standard accuracy remains consistently high. This confirms that, in the presence of class imbalance, accuracy alone is an insufficient evaluation metric.

As more samples from class 0 are included in the test set, the recall on that class decreases from 0.50 to 0.33 and finally to 0.25, highlighting the increasing difficulty the model faces in correctly identifying altered fertility cases.

A `test.size` of 0.25 was ultimately selected as it offers a good trade-off between the amount of data available for training and the reliability of the evaluation.

7 Conclusion

This work presents a machine learning system for male fertility diagnosis based on the Fertility dataset. Four classification models were evaluated using Balanced Accuracy as the primary metric.

The Support Vector Machine with an RBF kernel proved to be the best-performing model in cross-validation, achieving a Balanced Accuracy of 0.624. On the held-out test set, the model attained a standard Accuracy of 0.88 and a Balanced Accuracy of 0.644.

The results demonstrate that it is possible to achieve above-chance performance even on datasets of very limited size, although the obtained metrics remain insufficient for reliable application in a clinical setting. A larger and more balanced dataset would be necessary to develop a diagnostically viable classifier.